Compositions of representable functions are representable in Q

The Open Logic Project contributors

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Editorial scope. Statement and complete author proof are included verbatim as the span lines 1115-1132 of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the theory Q and its provability relation
- the representation formalism A_f for a function f in Q
- composition h = f after g of number-theoretic functions

Proposition 0.1. *If $h(n) = m$, then* $\mathrm{Th}Q \vdash !A_h(\underline{n}, \underline{m})$.

Proof. Suppose $h(n) = m$, i.e., $f(g(n)) = m$. Let $k = g(n)$. Then

$$\mathrm{Th}Q \vdash !A_g(\underline{n}, \underline{k})$$

since $!A_g$ represents g, and

$$\mathrm{Th}Q \vdash !A_f(\underline{k}, \underline{m})$$

since $!A_f$ represents f. Thus,

$$\mathrm{Th}Q \vdash !A_g(\underline{n}, \underline{k}) \wedge !A_f(\underline{k}, \underline{m})$$

and consequently also

$$\mathrm{Th}Q \vdash \exists[y][(!A_g(\underline{n}, y) \wedge !A_f(y, \underline{m}))],$$

i.e., $\mathrm{Th}Q \vdash !A_h(\underline{n}, \underline{m})$. □

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